Enhancing Battery Thermal Management With Virtual Temperature Sensor Using Hybrid CNN-LSTM

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***Abstract*— Temperature has a significant impact on lithium-ion batteries (LIBs) in terms of performance, safety, and longevity. A battery thermal management system is employed to ensure the safe operation of the batteries, especially during fast charging, high power discharge, and extreme weather conditions, thus enhancing their performance and prolonging their lifespan. The thermal performance of batteries is typically monitored using temperature sensors, which directly measure their surface temperature (ST). But, as a battery pack's number of cells increases, so does its number of temperature sensors, which raises its cost and reduces its reliability. To address this problem, this article introduces an innovative hybrid method leveraging deep learning algorithm, to accurately estimate the ST of LIBs. The methodology integrates the convolutional neural network (CNN), long short-term memory (LSTM), and deep neural network (DNN) components. Two distinctive CNN-LSTM configurations, series and parallel, are proposed for battery ST estimation. The effectiveness of the proposed approach is comprehensively validated using three distinct datasets with different chemistries and working operations. The validation process involves testing the model under two elevated ambient temperature of temperatures using constant current (CC) and Artemis urban drive profiles and on a battery subjected to various dynamic driving profiles across a range of ambient temperatures (10 °C–25 °C and −20 °C–10 °C). The accuracy of the estimation is assessed through root-mean-square error (RMSE), revealing an error of less than 1.24 °C and 1.30 °C for fixed and varying ambient temperature conditions, respectively, which demonstrate the robustness and reliability of the proposed hybrid approaches in battery ST.**

***Index Terms*— Battery temperature estimation, convolutional neural network (CNN), lithium-ion batteries (LIBs), long short-term memory (LSTM), parallel CNN-LSTM, series CNN-LSTM, surface temperature (ST).**

NOMENCLATURE

1D-CNN	One-dimensional convolutional neural network.
BMS	Battery management system.
CNN	Convolutional neural network.
DNN	Deep neural network.
EVs	Electric vehicles.
FNN	Feedforward neural network.
GRU	Gated recurrent unit.
LIBs	Lithium-ion batteries.
LSTM	Long short-term memory.
MAE	Mean absolute error.
MAXE	Maximum estimation error.
P CNN-LSTM	Parallel CNN-LSTM.
RMSE	Root-mean-square error.
RNN	Recurrent neural network.
S CNN-LSTM	Series CNN-LSTM.
SOC	State of charge.
SOH	State of health.
ST	Surface temperature.

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I. INTRODUCTION

A. Motivations

TEMPERATURE monitoring is a crucial task in BMS to ensure the safety, performance and reliability of battery operation in EVs [1]. Low temperatures can corrode battery electrodes and reduce battery performance, whereas high temperatures can decompose electrodes [2]. The high rate of charging and discharging or the long continuous operation of the LIBs can lead to high levels of heat being generated. As a result, the temperature can rise significantly, resulting in reduced battery performance, shorter battery life, and potentially even thermal runaway, which ends in destroying the battery. The temperature of a battery cell can be directly measured in a BMS using temperature sensors such as thermocouples and thermo-resistive sensors; however, due to several limitations, battery temperature estimation has become increasingly popular in recent years as a dependable and cost-effective alternative to sensor measurements. The significant limitation is its accuracy in extreme temperatures, which may result in inaccurate readings. Because of the slow response time of the sensors, the sensor may be unable to detect sudden fluctuations in battery temperature; furthermore, the sensors' durability should be considered since harsh conditions can shorten their lifecycle. Aside from the technical challenges involved with integrating the sensors into a battery pack, which makes it more complex and less reliable, another concern arises from the cost of adding these numerous sensors. It should also be noted that the placement and accuracy of temperature sensors can also be affected by the size, shape,

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and design of the battery. Regular calibration is, furthermore, essential to maintaining sensors' accuracy, which also makes regular maintenance necessary.

Temperature stands out as a primary factor impacting both the safety and performance of LIBs. It also plays a crucial role in influencing their degradation rate and remaining useful life (RUL). With the increasing integration of LIBs in EVs, there is a growing demand for faster charging and discharging strategies, contributing to a notable rise in cell Physically measuring the ST of individual cells poses challenges, including significant instrumentation costs and complexities related to the placement of sensors within the battery pack's hardware and wiring. To overcome these challenges, various methods have been proposed to determine the temperature of battery cells without requiring a dedicated sensor.

Broadly, there are some primary methods employed for battery temperature estimation. Physics-based thermal models leverage fundamental principles of heat transfer and thermodynamics to simulate the thermal behavior of a battery. They consider factors such as internal heat generation (HG), thermal conductivity, and convective heat transfer to predict temperature changes. Electro-thermal models integrate electrical and thermal equations to comprehensively simulate the interaction between a battery's electrical performance and thermal behavior. Data-driven approaches involve using machine learning (ML) techniques, such as regression models, neural networks, and predictive analytics. These methods are trained using historical temperature data to predict future temperatures, proving particularly valuable when the relationship between different parameters and temperature is intricate and nonlinear. Kalman filtering, a recursive algorithm, combines sensor measurements with predictions from a dynamic model. It is effective in tracking and estimating the state of a dynamic system, making it beneficial for real-time battery temperature estimation. Electrochemical impedance spectroscopy (EIS), commonly used in SOH estimation, can also monitor temperature changes within a battery. EIS-based methods leverage variations in impedance with temperature to gain insights into thermal behavior.

B. Research Review

Temperature estimation technology has been proposed to address the limitation of surface-mounted or intrusive sensors. One alternative solution is to use partial differential equations (PDE) thermal-based methods. In [3], for the determination of the battery's internal temperature, a lumped thermal model is developed based on the battery's voltage, current, and ambient temperature. In [4], a battery thermal model and Kalman filter were implemented to estimate battery temperature. A multistate model developed with PDEs provides an accurate description of battery thermal behavior; however, they are complex and require a comprehensive understanding of thermal characterizations of cells, thermal boundary conditions, and rates of HG, making them difficult to implement in real-time building management systems. The ST of batteries can also be estimated using EIS [5], [6], whereby a small ac current is injected into the battery, and its voltage response is observed to determine the relationship between its impedance and its temperature. Its implementation, however, usually increases the complexity of the BMS hardware and is not feasible for real-time operation [7].

The growing computing power and abundant data availability have paved the way for the increased application of ML methodologies in the realm of batteries. These methodologies establish connections between measured data and crucial battery states such as SOH and SOC throughout their lifecycle. Previous studies have predominantly focused on ML strategies for estimating battery SOC [8], [9], [10], [11], and SOH [12], [13], [14], employing internal nonlinear time-variant battery models derived from historical data. A technique for predicting HG in LIBs using a shallow neural network is introduced in [15]. This model was evaluated for its ability to simulate thermal management system control and detect sensor faults.

To date, only a small number of studies have investigated the effectiveness of deep learning methods in estimating battery ST. Hussein and Chehade [16], introduced models based on FNN and RNN that use measured voltages and currents, along with the previously estimated temperature, to predict battery STs accurately. The investigation, however, focused solely on temperatures of 25 °C, raising questions about the model's performance under a broader range of conditions. In [17], the internal core temperature of a battery is estimated through the use of a radial basis function FNN model and a Kalman filter. For estimating internal core temperature, the model is based on measured ST, voltage, and current. Despite the excellent results reported, since measured STs are inputs to the model, they cannot be compared with algorithms that aim to estimate STs. Temperature estimation is a time series distribution; however, the standard RNN suffers from the gradient vanishing or rare gradient explosion phenomena, which prevents the detection of long-term dependencies. More advanced structures have, therefore, been proposed to address this issue by introducing DNN [18], including more hidden layers with deeper fully connected (FC) nodes to enable the network to process long sequences of data. For instance, in the research outlined in [20], an investigation was conducted into the accuracy of FNN and LSTM models in estimating battery STs at different operations by using current, voltage, ambient temperature, and SOC. In [19], a GRU RNN model was developed to estimate battery ST using voltage, current, SOC, and ambient temperature as the input. The model provided accurate results but was only validated on a limited range (varying temperature of 10 °C to 25 °C).

C. Contributions

While previous studies have explored the application of DNNs for temperature estimation, their investigation has been limited to assessing a comprehensive range of operating conditions; moreover, despite the recognized effectiveness of hybrid DNN networks in estimating battery SOC and SOH [20], [21], [22], there remains a notable research gap in the exploration of hybrid models in the domain of battery ST estimation. In response to this gap, our study is dedicated to advancing this field by introducing hybrid deep learning approaches.

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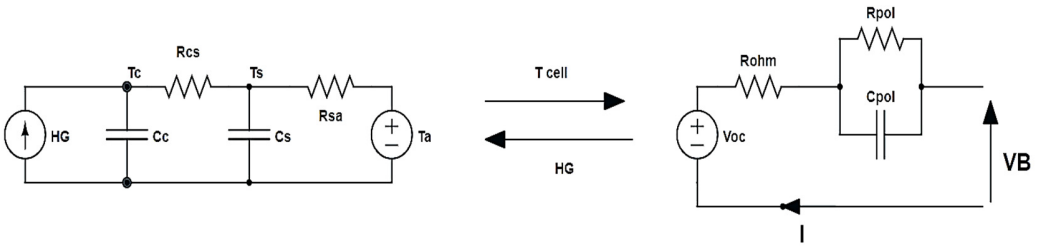


Fig. 1. Battery equivalent circuit thermal model and battery equivalent circuit electric model.

An initial attempt has been made to enhance the accuracy and adaptability of battery ST estimation with two novel hybrid models, S CNN-LSTM and P CNN-LSTM.

Our primary contributions can be summarized as follows:

1) This study introduces two novel hybrid models–the S CNN-LSTM and P CNN-LSTM–marking a pioneering effort to elevate precision and adaptability in battery ST estimation. The fusion of CNN and LSTM structures represents a key innovation, surpassing current methodologies and addressing the highlighted research gap.

2) A rigorous evaluation of our proposed hybrid models was conducted under various realistic and challenging operating conditions, taking different types and chemistries of batteries into account, to ensure their practical applicability. This encompassed challenges such as low ambient temperature, high-temperature rise, fast charging, and aging effects.

3) Beyond model development and validation, we provide a distinctive and explicit analysis of the proposed hybrid models. This involves a comprehensive benchmarking against prior studies in the literature, offering a detailed comparison that underscores the uniqueness and effectiveness of our approach in the context of battery ST estimation.

Following is a summary of the rest of this article. Section II presents an overview of the temporal patterns inherent in the distribution of battery temperatures. Section III introduces the architecture of CNN, LSTM, and DNN. In Section IV, we offer comprehensive descriptions of three datasets used in our study. In Section V, our innovative approach, leveraging the CNN-LSTM model for battery temperature prediction, is illustrated. Section VI conducts a meticulous analysis of our model's results, providing a deep understanding of their performance. Finally, we conclude our article in Section VII, summarizing our key findings and emphasizing their significance.

II. Battery Temperature Analysis

The intricate and nonlinear thermal behavior of the battery necessitates a comprehensive analysis of data over time to determine their ST. Battery temperature is intricately linked to its HG, which occurs through both reversible and irreversible mechanisms during the processes of charging and discharging. Irreversible heat is generated by the joule effect as electric current passes through the battery cells, and reversible heat arises from polarization in the electrochemical reactions at the cathode and anode. Heat transfer involving conduction, convection, and radiation contributes to an increase in temperature within the battery cells as heat accumulates. The HG in the normal operation of a LIB is typically expressed by the following simplified equation, derived from [23]:

$$\mathrm{HG} = I(V_B - V_{ocv}) - IT\frac{\partial V_{ocv}}{\partial T}. \tag{1}$$

In this equation, the first term represents heat generated by the battery's joule resistance (ohmic resistance), while the second term signifies reversible HG or depletion resulting from bilateral entropy exchange during the electrochemical reaction [24]. The variables I, V_B, V_{oc}, and T denote current, terminal voltage, open circuit voltage and temperature. Generally, the heat generated or lost due to reversible entropy exchange is negligible for EV current rates, given the dominance of ohmic or joule heating. Presently, the widely adopted method involves using the simplified equation, focusing solely on the first term, $I(V_B - V_{ocv})$. Both I and V_B representing current and terminal voltage can be directly measured by physical sensors. The battery's internal equilibrium voltage or V_{ocv} is a nonlinear function, which directly related to its SOC. Hence, the SOC is a suitable representation of V_{ocv} for heat analysis. Thermal analysis can be streamlined by incorporating two assumptions: 1) the generated heat is uniformly distributed, and 2) the battery surface maintains a uniform temperature. Fig. 1 illustrates a simplified thermal model of a battery's equivalent circuit, which can describe the correlation between HG and battery temperature

$$\begin{aligned} C_C\frac{dT_C}{dt} &= \mathrm{HG} - \frac{T_c - T_s}{R_{cs}} \\ C_S\frac{dT_S}{dt} &= \frac{T_c - T_s}{R_{cs}} - \frac{T_s - T_a}{R_{sa}}. \end{aligned} \tag{2}$$

Here, C_C and C_s denote the core and surface heat capacity of the battery, and T_c and T_s represent the core and surface battery temperature. T_a corresponds to the ambient temperature, while R_{cs} and R_{sa} are the resistances of the core to surface and surface to ambient, respectively. To provide insight into the time series nature of ST distributions, (1) and (2) are expressed as state functions in continuous time dynamic systems

$$\begin{bmatrix} \dot{T}_C \\ \dot{T}_S \end{bmatrix} = \begin{bmatrix} -\frac{1}{C_C R_{cs}} & \frac{1}{C_C R_{cs}} \\ \frac{1}{C_c R_{cs}} & -\frac{2}{C_c R_{cs}} \end{bmatrix} \begin{bmatrix} T_C \\ T_s \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{C_C} \\ \frac{1}{C_s R_{sa}} & 0 \end{bmatrix} \begin{bmatrix} T_a \\ I(V_B - V_{ocv}) \end{bmatrix}. \tag{3}$$

In reference to these state equations, the ST of the subsequent time step ($\dot{T}_S$) is influenced by the battery's inherent chemical properties (C_C, C_s, R_{cs}, and R_{sa}), HG and ambient

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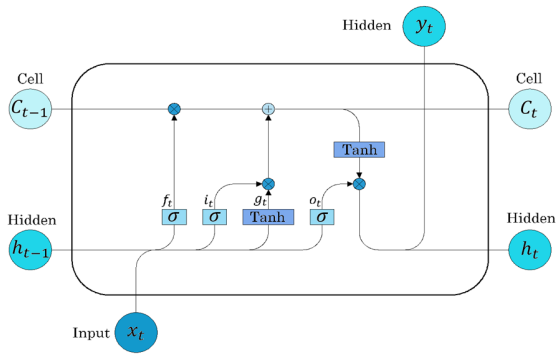


Fig. 2. Internal structure of LSTM block.

temperature (HG, and T_a), and the current ST (T_s). Clearly, this inspection illustrates that the task of predicting battery ST is inherently time series in nature.

III. Methodologies

In this section, an overview of deep learning methods used in our study is provided. Then the proposed model will be described later. Generally, the procedure includes two phases: the model training phase and the prediction of real-time battery ST. The data preprocessing phases, which include data normalization, are central to all these phases. The proposed predictive model is first developed during the model training stage. It is fine-tuned by using historical data collected from battery operation experiments as a training dataset. The next step is real-time analysis of the ST prediction model, which enables us to make well-informed decisions regarding maintenance strategies and overall asset management based on current operational data.

A. Data Normalization

The battery dataset contains a different range of sensor measurement degradation data with a variety of units. Data with small values may easily lose their features without data preprocessing, so it is necessary to perform standard data normalization to make training faster and more accurate. By using zero normalization, the data is standardized. To normalize a Z-Score, the standard deviation and mean of each attribute within a training set are calculated, which serve as a reference. Normalization of the Z-Score follows the following formula:

$$x_n^i = \frac{x^i - \mu^i}{\sigma^i} \tag{4}$$

where each observed instance, x^i, is transformed into x_n^i for the ith feature in the history data. μ^i and σ^i are the mean and variance of the ith measured feature, respectively.

B. Long Short-Term Memory

LSTM is an RNN developed to solve gradient vanishing or exploding problems in standard RNN [25]. An LSTM consists of three inputs (h_{t-1}, C_{t-1}, and x_t) and two outputs (h_t and C_t), along with three types of gates: input gate, forget gate, and output gate, Fig. 2.

The forget gate f_t is responsible for deciding on the information from the preceding cells that should be allowed to pass and information that should be suppressed. Equation (5) outlines the definition of the f_t, when it takes inputs from the prior hidden layer state, h_{t-1}, and the current input sequence x_t at time t and produces an output ranging between 0 and 1. A value of 1 signifies the preservation of all previous information, while a value of 0 indicates that nothing is retained from the preceding cell state. This mechanism empowers the LSTM to effectively regulate the flow of information, preventing the loss of crucial data in time series analysis

$$f_t = \sigma\left(W_f.\left[h_{t-1}, x_t\right] + b_f\right). \tag{5}$$

Here f_t represents the state of the forget gate at time t; σ is a sigmoid activation function controlling information flow; W_f is the weight; h_{t-1} refers to the output value at time $t-1$, x_t represents the input value, and b_f stands for the bias term.

Following this step, the input gate determines the new data that should be merged into the cell state through a two-step process. As a start, input gate layer determines the values that need updating; then, new values can be added to the state using the hyperbolic tangent (tanh) function. Through the joint action of the first three outputs (f_t, i_t, g_t) the state of the memory cell in the previous stage C_{t-1} is updated to state C_t. These processes are described by the equations below

$$i_t = \sigma\left(W_i.\left[h_{t-1}, x_t\right] + b_i\right) \tag{6}$$

$$g_t = \tanh\left(W_g.\left[h_{t-1}, x_t\right] + b_c\right) \tag{7}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot g_t. \tag{8}$$

Here i_t stands for the input gate at time t, which regulates the flow of information from x_t to g_t. The input gate's weights and biases are denoted as W_i and b_i, respectively. g_t represents candidate memory cell, which is determined by tanh activation function using W_g and b_c correspond to the weight matrix and bias term of the cell state. The previous memory C_{t-1} is updated based on the information generated in the previous two steps and renamed to C_t through the Hadamard product, $\odot$, element-wise multiplication.

The output gate, O_t, plays a crucial role in selecting relevant information from the current cell state for output. Additionally, a hyperbolic tangent layer processes the chosen information, generating the ultimate output value. This output value is subsequently forwarded as input to the succeeding hidden layer within the network. Equations (9) and (10) describe how the output gate's activation is calculated

$$O_t = \sigma\left(W_o.\left[h_{t-1}, x_t\right] + b_o\right) \tag{9}$$

$$y_t = h_t = O_t.\tanh(C_t). \tag{10}$$

Here O_t represents the output gate, regulated by a sigmoid layer denoted as σ using the weight matrix W_o and bias term b_o, to determine the portions of the memory cell state to be exported. The hidden state at the current time, denoted as h_t or y_t, serves as the ultimate output of the LSTM unit. This output is derived from the product of the activated output data O_t through the tanh function and the new state C_t, ultimately forming the output of the LSTM unit.

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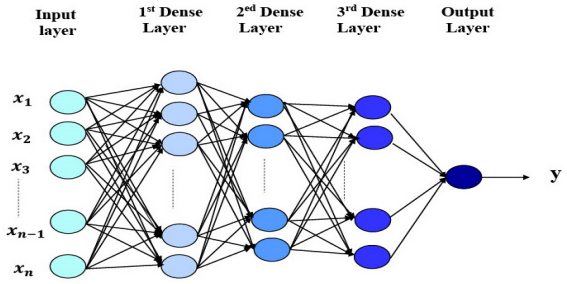


Fig. 3. Internal structure of DNN block.

C. Deep Neural Network

ML has been revolutionized by the advent of DNNs, which have achieved state-of-the-art results in numerous applications. DNNs are from the FNN family and can learn highly nonlinear and complex representations of input data, which is one of the primary reasons why they perform better than traditional ML algorithms. Nevertheless, they have gained popularity as hardware and software technologies have advanced.

DNNs are composed of multiple hidden layers, allowing for the processing and transformation of complex input data through multiple stages of abstraction. The mathematical form of the mapping in DNN can be expressed as a composition of functions across the layers as follows:

$$O_{\text{DNN}} = f^{(l)}\big(f^{(l-1)}\big)\big(\ldots f^{(2)}\big(f^{(1)}\big(\text{X}W^{(1)} + b^{(1)}\big)W^{(2)} + b^{(2)}\big)\ldots\big)\ W^{(l-1)} + b^{(l-1)}\big)W^{(l)} + b^{(l)}\big)\big). \quad (11)$$

Here O_{DNN} stands for the output, $f^{(l)}(\cdot)$ represents the activation function applied elementwise at layer l, which is sigmoid in the proposed model. $W^{(l)}$ and $b^{(l)}$ are the weight matrix and bias vector at layer l. X refers to the input ad l is the total number of layers in the network. The structure of DNN with three hidden layers is demonstrated in Fig. 3.

D. Convolutional Neural Network

CNN are FNNs presented by [26] for feature extraction and pattern recognition. As a powerful image processing technology, CNNs are well known in 2-D formats. There is a 1-D form of CNN, 1D-CNN, that can be used for the processing of data with temporal characteristics. Several investigations have been conducted on the application of CNNs to deal with large amounts of time series forecasting and estimation and fault diagnosis because of their advantage in automatic feature extraction and low risk of overfitting.

CNNs are a type of multilayer perceptron (MLP) comprised of an input, hidden, and output layer. The input layer transfers the original data to the hidden layer. The output layer has the task of providing specified outputs. The convolutional layer, pooling layer, and FC layers are considered as hidden layers. The sparse connectivity, shared weights, and pooling architecture of a CNN result in significantly fewer parameters needed to maintain accuracy compared to traditional DNNs. CNNs use convolutional filters with shared weights, meaning the same filter is applied to multiple regions within the input data. The sparse connectivity of a neural network can be defined as the use of a limited number of connections between neurons in adjacent layers, which means the neurons in one layer are only connected to a subset of neurons in the previous layer. With a pooling architecture, the outputs of convolutional layers are replaced with summary statistics to ensure that the output is not sensitive to changes in the input. The use of all three features results in a network with fewer parameters, which improves computation efficiency and prevents overfitting.

The core of a CNN lies in its convolutional layers, which collect local characteristics from inputs of upper layers through convolution operations. Subsequently, these layers transmit all acquired information to lower levels to capture more intricate features. The convolutional operation is defined as below

$$c[n] = x[n] * k[n] = \sum_{m=0}^{l-1} x[m].k[n-m] \quad (12)$$

where $k[n]$ represents the convolution kernel with size l, and $c[n]$ represents the convolution output.

The convolutional layer is comprised of filter kernels that share identical weights. In a CNN, this layer operates by sliding a filter matrix (or kernel) across the input data, conducting element-wise multiplication and accumulation at each region, and subsequently incorporating a bias term. The output vector of the lth convolutional layer is expressed as [27]

$$x_j^l = \sigma\left(b_j^l + \sum_{i=1}^{m^{l-1}} w_{j,i}^l x_i^{l-1}\right)\big(j = 1, 2, \ldots, , m^l\big) \quad (13)$$

where x_j^l demonstrates the jth output feature on the lth layer; σ, b_j^l, and $w_{j,i}^l$ indicate the sigmoid activation function, the bias for the jth feature map, and the weight vector corresponds to the convolutional kernel between the ith feature on the lth layer and the jth feature on the $(l-1)$th layer. x_i^{l-1} is the ith generated features by the previous layer $(l-1)$, m^l is the number of features on the lth layer.

Following that, robust local features are extracted from pooling layers by applying a pooling operation, such as max pooling or average pooling, to a region of the feature map. The pooling layer is usually inserted between two convolutional layers and linked to the feature map from the preceding convolutional layer. As a result, it can reduce the computational power required to process the data while still allowing its dominant features to be extracted from feature maps, which reduces the resolution of the feature maps and avoids network overfitting, and thus, making it easier to train the network.

$$y_p^l = f\big(y_c^{l-1}\big) = \underbrace{\max}_{\forall c \epsilon p_s}\big\{y_c^{l-1}\big\}. \quad (14)$$

Given the better performance of max pooling in our proposed model, feature maps generated from the conv1D layers are given as inputs to retain only the maximum in a region of the feature map. The window is moved across the feature map with a specific stride, which operates based on the following equations, respectively: where max operation is the downsampling function, y_c^{l-1} is the input vector from the previous layer and y_p^l is the output feature vector, P_S represents the pooling size, which is a hyperparameter.

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TABLE I
DATASET AND THEIR TEST SPECIFICATIONS

Dataset	Type	**Material**	Voltage	Capacity	Charge Current	Discharge current	Ambient Temperature
Dataset1	Cylindrical	LiNiCoAlO2 (NCA)	2.5-4.2V	2 Ah	1.5 A	CC at 4A	43°C
Dataset2	Pouch cell	LiNiMnCoO2 (NMC)	2.7-4.2V	0.74 Ah	1.48A	Artemis Urban drive cycle	40°C
Dataset3	Cylindrical	LiNiCoAlO2 (NCA)	2.7-4.2V	2.9 Ah	2.9A	UDDS, HWFET, US06, LA92, Mixed	Constant -20, -10, 0, 10and 25 ºC Varying from -20 ºC to 25 ºC

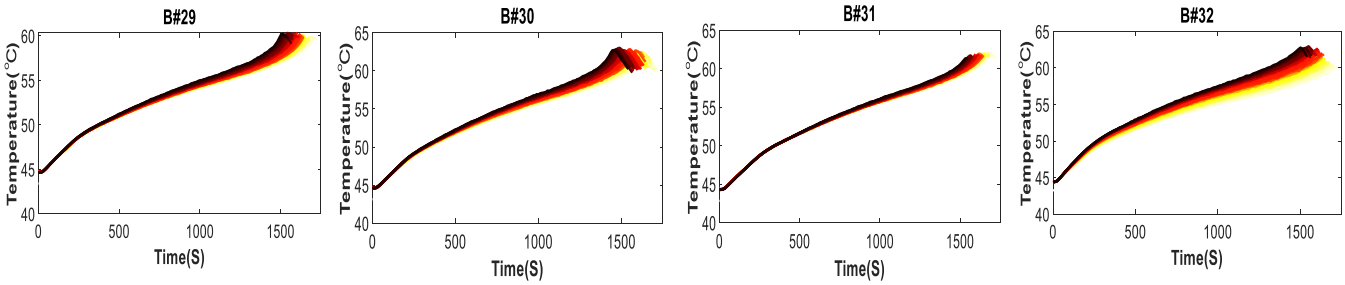


Fig. 4. NASA batteries ST through the discharge cycle.

The pooling operation applies to nonoverlapping regions of the input data, the window slides over the input data, and output is the total value of each region. FC layers, also known as dense layers, map the extracted features into a final output to be used in further classification or regression analyses. Typically, this layer is placed at the end of a CNN architecture, in which each neuron of this layer is connected to all neurons in the previous layer.

IV. DESCRIPTION OF DATASETS

It is imperative to confirm that battery temperature can be accurately assessed for different battery chemistries and operating conditions. To address this, we acquired three prominent open-source datasets related to diverse battery types tested across various operational scenarios. Comprehensive specifications for each dataset are outlined in Table I. These datasets were sourced from NASA, Oxford University, and the University of Wisconsin–Madison. The NASA dataset captures data at steady temperatures during constant current (CC) discharge cycles, while the Oxford University dataset records data under stable temperature conditions under dynamic driving cycles. The University of Wisconsin–Madison dataset encompasses data collected across both stable and fluctuating temperature settings. These datasets serve as validation sources for our model.

A. Dataset 1: NASA Dataset

A study conducted by the NASA Prognostics Center of Excellence examined 18 650 Li-ion batteries with a rated capacity of 2 Ah [28]. In a series of battery experiments, different sets of Li-ion batteries were subjected to various

operational profiles, including charging, discharging, and impedance measurements, each set experiencing unique conditions. Four Li-ion batteries (# 29, 30, 31, and 32) were subjected to the three operational profiles but at an elevated ambient temperature of 43 °C. Charging was performed in two stages: CC mode at 1.5 A until the battery voltage reached 4.2 V, followed by constant voltage (CV) mode until the charge current dropped to 20 mA. Discharge was carried out at specific current levels until battery voltage thresholds were met, varying for each battery. The experiments continued until the batteries reached the end-of-life (EOL) criteria, marked by a 30% decrease in rated capacity, which was 2 Ah reduced to 1.4 Ah. Fig. 4 illustrates the batteries' ST over the entire discharging cycle, where the color bar shows the new brand battery over its aging. For training the model on Dataset 1, the NASA dataset, three batteries (B#30, B#31, B#32) were used. For testing the proposed model, B#29 was isolated from the dataset.

B. Dataset 2: Oxford Dataset

The second dataset employed in this study comprises eight Kokam pouch cells, with a capacity of 740 mAh and a nominal voltage of 4.2 V, subjected to an aging test within a controlled environment set at 40 °C. The aging protocol consisted of two distinct phases: characterization tests and drive cycles. Characterization tests were conducted at regular intervals of every 100 driving cycles. During these tests, the cells underwent the charging process at a CC of 740 mA (1-C) until reaching a voltage of 4.2 V, followed by discharge at the same current until the voltage dropped to 2.7 V. The drive cycles involved a 2-C charge cycle and dynamic discharge operation, emulating the real-world driving scenarios using the Artemis urban drive cycle. The criteria for ending the characterization test were established to stop once the battery's capacity had reduced to roughly 80% of its designated capacity. Measurements of the battery were taken every second. For additional details on the experimental arrangement, readers are directed to [29]. Figs. 5 and 6 demonstrate all cells' ST measurements and batteries degradation over their lifecycles, respectively.

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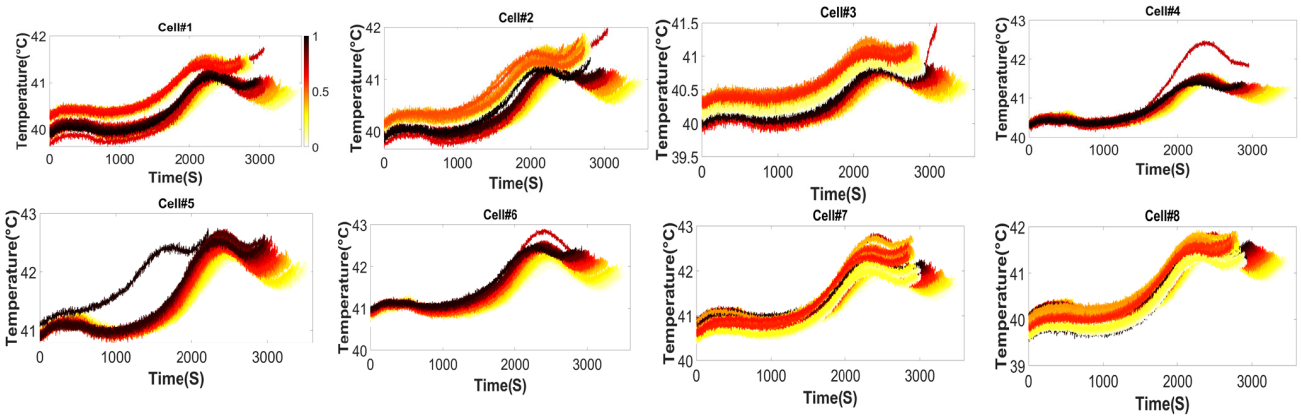


Fig. 5. Oxford eight experimental LIBs ST through discharge cycle.

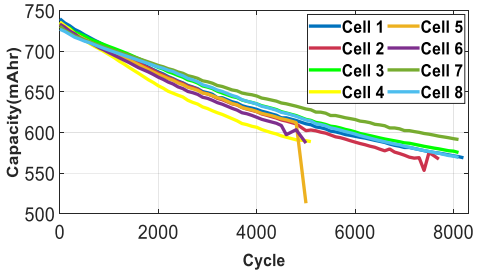


Fig. 6. Capacity degradation evolution of eight experimental LIBs.

Despite having identical operating conditions, the aging patterns differed among the batteries. Cell 5 displayed premature failure at around 5000 cycles, whereas Cells 2, 4, and 6 underwent whereas Cells 2, 4, and 6 underwent sudden regime changes at different cycle counts. The input parameters of the model comprised voltage, current, ambient temperature, accumulated charge, and their respective moving averages, with the goal of validating the practical applicability of the model. Temperatures recorded around 0 °C in a single cycle were excluded from the analysis due to their notable deviation from the target temperature of 40 °C. To train the models, Cells 1, 2, 3, 4, 6, 7, and 8 were selected. Conversely, Cell 5, distinguished by its premature failure and distinct temperature pattern compared to other cycles, was reserved for testing. Data observations within a 30-s timeframe are used for both training and testing the model.

C. Dataset 3: University of Wisconsin–Madison and McMaster University Dataset

Under fixed and varying temperatures, University of Wisconsin–Madison researchers assessed a Panasonic 18650PF Li-ion battery cell with Lithium Nickel Cobalt Aluminum Oxide (LiNiCoAlO2) chemistry and 2.9 Ah rated capacity [30]. The Tests were conducted at fixed and different ambient temperatures, ranging from −20 °C to 25 °C. A heating and cooling element was used to control the air temperature during the tests, and the chamber fans were always on. The cylindrical housing of the cell was equipped with a temperature sensor, and an air temperature sensor was placed inside to measure air temperature. As part of the fixed temperature tests, the ambient temperature was −20 °C, −10 °C, 0 °C, 10 °C, and 25 °C, whereas varied ambient temperature tests were run at either −20 °C or 10 °C, with the cooling system turned off but the circulating fans running. In these tests, heat transfers through the chamber walls result in a natural rise in chamber temperature, reaching 35 °C. Each test begins with a fully charged battery cell at 20 °C, followed by a power drive cycle profile until battery SOC reaches an end value of between 10% and 30%, at which point the battery is incapable of providing enough power to continue driving the vehicle.

The driving cycle power profile was derived from a 35 kWh battery pack model for an electric Ford F150 truck and scaled up for a single 18650PF cell. The battery is cycled using four standard drives and five mix drives, which are composed of randomized mixtures of the standard cycles' power profiles. These standard cycles comprising Urban Dynamometer Driving Schedule (UDDS), Los Angeles 92 (LA92), Supplemental Federal Test Procedures (US06), and Highway Fuel Economy Driving Schedule (HWFET) with maximum charge and discharge currents at 25 °C are 4.2/7.5, 9.7/10.5,7.5/20.8, and 5.5/5.5 A, respectively. The five mixed drive cycles consist of a randomized combination of standard drive profiles, referred to as Cycle1∼Cycle 4 and NN drive cycle. The drive cycle continued until the battery SOC reached an end value of between 10% and 30%, when the battery was unable to provide enough power to continue. Based on the practical limitations of a battery management system, the original dataset was downsampled to 0.1 Hz. Fig. 7 provides a visual depiction of the changes in batteries' SOC (ST) across all various driving profiles, considering both fixed and varying temperature conditions. In the course of training the model on this dataset, our proposed models were trained using five mixed drive cycles, each conducted at a consistent ambient temperature, and these cycles were designated as the "Training data" in Table II. The "Validation" dataset comprised four standard drive cycles executed under stable ambient temperature conditions. To test the models, we employed mixed drive cycles that featured varying ambient temperatures. We resampled both the training and testing datasets to a frequency of 0.1 Hz.

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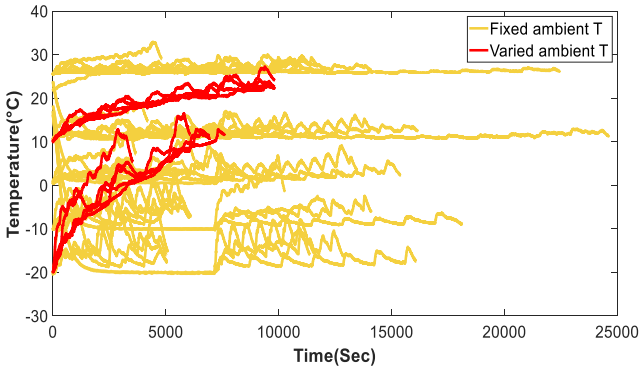


Fig. 7. Batteries ST through discharge cycle.

TABLE II
UNIVERSITY OF WISCONSIN–MADISON AND MCMASTEREXPERIMENT SPECIFICATIONS

Ambient Temperature	Drive cycle	Usage
Fixed Temperature at -20, -10, 0, 10, 25 ºC	Cycle1, Cycle2, Cycle3, Cycle4, NN	Training
Fixed Temperature at -20, -10, 0, 10, 25 ºC	UDDS, LA92, US06, HWFET, NN	Validation
Varying Temperature Start at -20ºC, 10 ºC	Cycle1, Cycle2, Cycle3, Cycle4	Test

V. PROPOSED BATTERY ST ESTIMATION

The initial prediction model was designed based on the LSTM architecture; however, better accuracy prompted us to embark on a transformative research path, which resulted in the development of two hybrid models leveraging LSTM, CNNs, and DNNs. With this fusion, we proposed S CNN-LSTM and P CNN-LSTM, an entirely new approach for ST estimation of a battery. These methods, furthermore, represent a significant advance in the field of battery state estimation by surpassing the limitations of the standalone LSTM model. The estimation accuracy of the LSTM model is also presented along with the other models to justify using more complex models. As described earlier, the original structures of the CNN and LSTM cells are retained in the proposed model. Generally, model prediction accuracy can be improved by adding more convolutional and recurrent layers to the models, but excessive layers may result in overfitting and unnecessary calculations. The LSTM model comprises six layers, while the hybrid models are designed to encompass thirteen layers. The detailed configurations of LSTM and hybrid models are illustrated in Figs. 8–10.

The S CNN-LSTM model comprises an initial input layer, followed by a convolutional layer, a nonlinear activation as a rectified linear unit (ReLU layer), a batch normalization layer, and a pooling layer. Subsequently, the features extracted by the CNN are fed into a two-layer LSTM recurrent network. To enhance the accuracy of estimation, a DNN model consisting of three dense hidden layers and a dropout layer

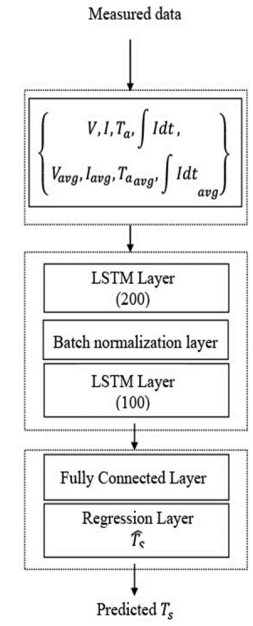


Fig. 8. Architecture of LSTM.

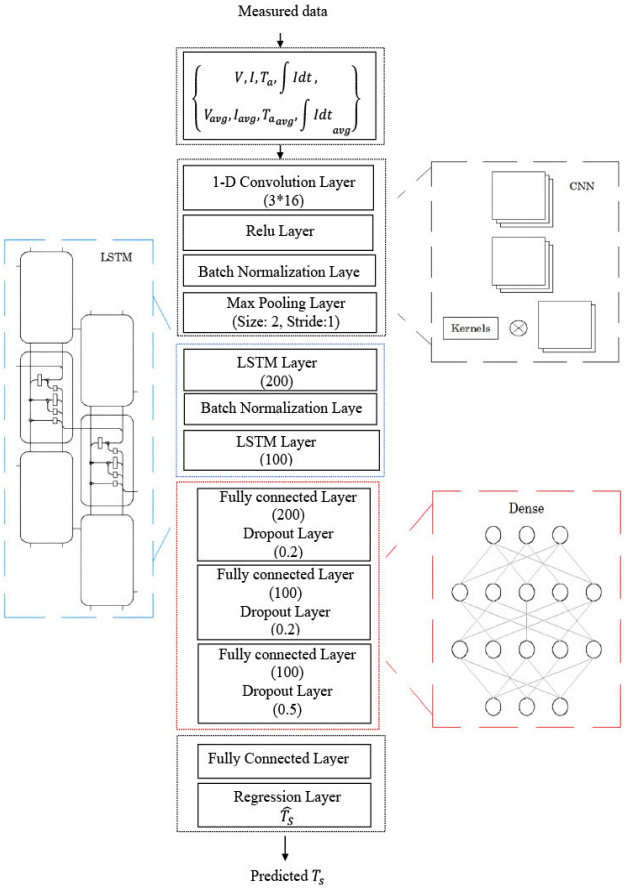


Fig. 9. Architecture of S CNN-LSTM.

is employed. Finally, an output layer is employed to generate the estimation.

The process commences with an input layer, which accepts sequential data, where battery variables, including voltage (V), current (I), ambient temperature(T_a), integration of current

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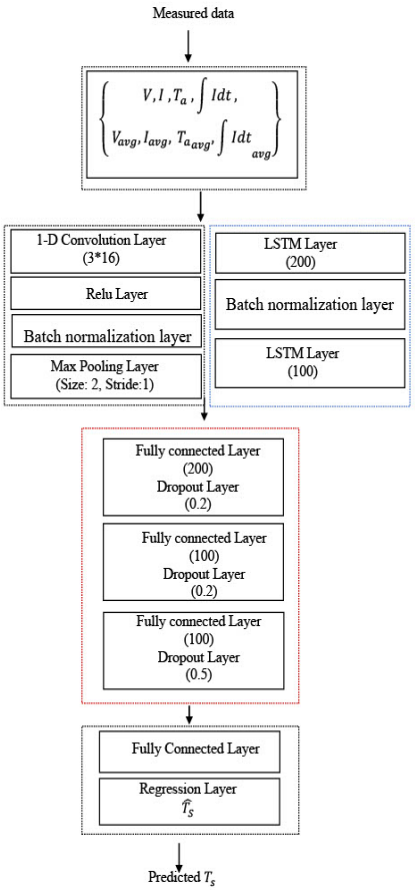


Fig. 10. Architecture of P CNN-LSTM.

over time and their moving average values are fed into the network. Specifically, the average values are moving averages referring to [31] over 30 precedent time steps, in which better performance was achieved when the average current and voltage were present. A 1-D convolutional layer with 16 filters of size 3 adeptly extracts pertinent features from the input data. Following this, the incorporation of a ReLU layer introduces nonlinearity into the model. To enhance the model's stability and training efficiency, a batch normalization layer is applied, providing normalization for improved convergence. Subsequently, a max-pooling layer with a pool size of 2 and a stride of 1 is employed to reduce spatial dimensions, effectively condensing the information while retaining crucial details. This step contributes to both computational efficiency and the extraction of key patterns in the data. The features extracted from the previous layers are then input into an LSTM model, which consists of two LSTM layers. Each LSTM layer is equipped with 200 units, serving to effectively capture and model the sequential nature inherent in the data. Then, three FC layers, featuring 200, 100, and 100 neurons, are added, which each followed by intermittent dropout layers for regularization (dropout rates of 0.2, 0.2, and 0.5), respectively. The final FC layer serves as the output layer, potentially used for regression tasks. This comprehensive architecture combines convolutional, recurrent, and FC layers with dropout regularization, offering a powerful framework for sequential data analysis in battery ST applications.

TABLE III
TRAINING PARAMETERS

Training Software	MATLAB 2023.a, Deep Learning Toolbox
Training Platform	Intel Core i7 CPU
Loss Function Optimizer	ADAM
Learning Rate	0.001
Number of Mini batches	Dataset1= 32, Dataset2=64, Dataset3=1
Validation Patience	20 epochs
Stopping Criteria	20 epochs with no loss improvement
# Training Repetitions	3-20

The P CNN-LSTM model, tailored for parallel feature processing, mirrors the structure of the S CNN-LSTM model in terms of number of neurons, layer configurations, and other specifications. In this parallel design, both the CNN and LSTM components individually extract features directly from the input data. The process begins with an input layer that accepts sequential data, including V, I, T_a, integration of current over time and their moving average values are fed into the network. 1-D convolutional layer with 16 filters of size 3 adeptly extracts pertinent features from the input data. Then, the incorporation of a ReLU layer infuses nonlinearity into the model. To enhance the model's stability and training efficiency, a batch normalization layer is applied. Subsequently, a max-pooling layer with a pool size of 2 and a stride of 1 reduces spatial dimensions, effectively condensing the information while retaining crucial details.

Two LSTM layers, each housing 200 units, are employed to model the sequential nature of the data. Subsequent to this, the extracted features from both the CNN and LSTM are concatenated and fed into three FC layers, featuring 200, 100, and 100 neurons, respectively. Each FC layer is succeeded by intermittent dropout layers for regularization, with dropout rates of 0.2, 0.2, and 0.5, respectively. The final FC layer serves as the output layer, suitable for regression tasks, serves as the output layer, suitable for regression tasks.

This parallel structure seamlessly integrates characteristics from both the convolutional and recurrent layers with dropout regularization, establishing a resilient framework for analyzing sequential data in applications related to battery ST.

To assess the performance of the suggested model, both models share identical LSTM structures. Parameters such as the learning rate, number of iterations, the quantity of LSTM cells, the size of hidden nodes, and the optimization algorithm are consistent when training both models on the given dataset. Nonetheless, there are variations in the batch size and validation frequency hyperparameters for datasets, as their data characteristics, size, and nature differ. It is worth noting that the model architecture remains unchanged. Further details on the model training process on datasets are provided in Table III.

VI. TEMPERATURE PREDICTION RESULTS AND DISCUSSIONS

In the S CNN-LSTM model, the CNN plays the pivotal role of extracting spatial features from sequential battery data. The CNN collectively serves to efficiently capture and highlight

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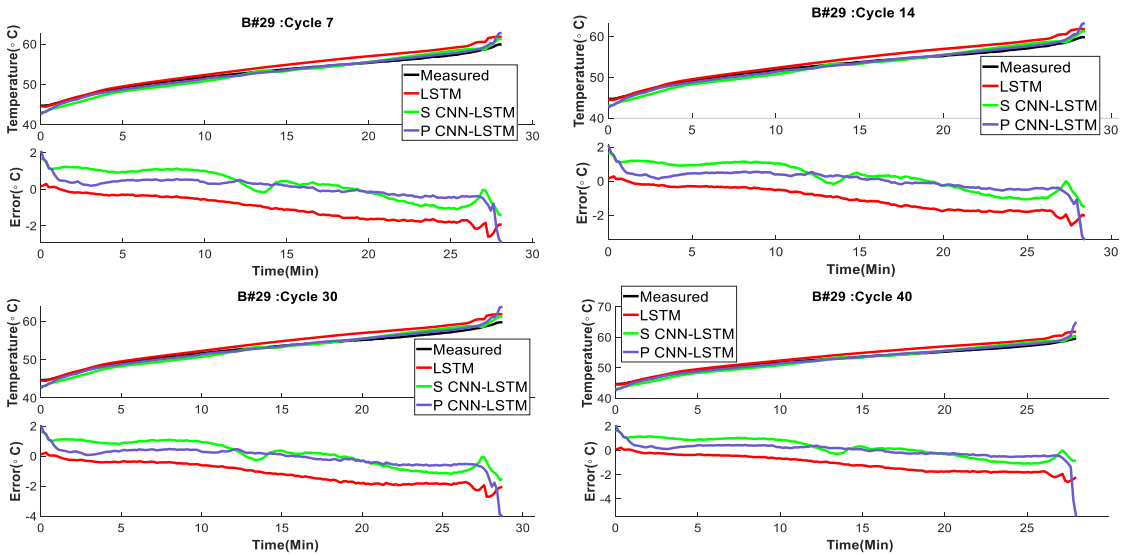


Fig. 11. ST estimation over cycles and its corresponding error for B#29 at a fixed temperature of 43 °C.

pertinent patterns in the input data. Subsequently, the features unearthed by CNN are seamlessly integrated into the LSTM. The LSTM is designed to capture and model the sequential dependencies inherent in extracted features. Lastly, a densely connected neural network (DNN) further refines the extracted features, enhancing the overall accuracy of the temperature estimation.

In the P CNN-LSTM model, designed for parallel feature processing, both the CNN and LSTM components autonomously extract features directly from the input data. The CNN functions as a dedicated spatial feature extractor, adeptly identifying pertinent patterns in sequential battery data. Concurrently, the LSTM component captures sequential dependencies, recognizing and retaining temporal patterns. Unlike the S CNN-LSTM, the parallel design ensures that spatial and temporal aspects are processed simultaneously, facilitating a comprehensive understanding of battery input data. This architecture leverages the strengths of both CNN and LSTM networks for feature extraction from input, offering a robust framework for accurate temperature estimation in battery surface applications.

A. Validation Against Manufacturing Uncertainty and Aging: Dataset 1

In this experiment, we assessed the model's accuracy in predicting the behavior of batteries' ST as they aged under elevated temperatures of 40 °C. Fig. 11 demonstrates the estimation results and their corresponding errors for some random cycles of test batteries labeled as B#29. Fig. 12 visually demonstrates all RMSEs and MAEs estimation of the three different models, LSTM, S CNN-LSTM, and P CNN-LSTM, across the battery's lifetime, which are also summarized in Table IV. In the case of LSTM, the RMSE falls within the range of 1.18 to 2.23, accompanied by an MAE spanning from 0.99 to 2.18. The S CNN-LSTM model exhibits a more constrained error range, with RMSE values

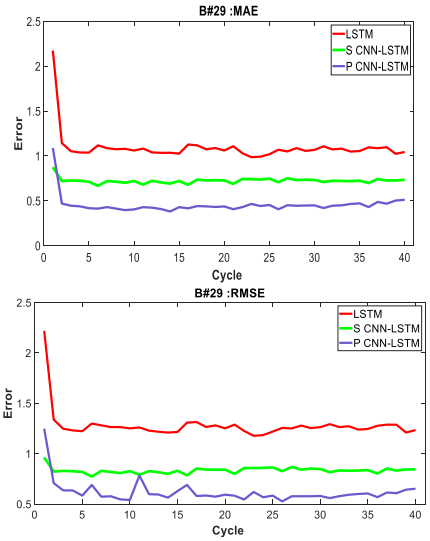


Fig. 12. RMSE and MAE of three models on battery #29 subjected to 43 °C ambient temperatures.

TABLE IV
RMSE AND MAE ST ESTIMATION ON B#29

Model	B#29	
	RMSE	MAE
LSTM	1.18 - 2.23	0.99 - 2.18
S CNN-LSTM	0.77 - 0.97	0.67 - 0.88
P CNN-LSTM	0.53 - 1.24	0.38 - 1.08

between 0.77 and 0.97, and MAE ranging from 0.67 to 0.88. For P CNN-LSTM, the RMSE is found to be between 0.53 and 1.24, with corresponding MAE values in the range of 0.38 to 1.08. The proposed model showed accurate estimations for the test batteries throughout their aging process when the battery's internal resistance increased.

These evaluation metrics provide a comprehensive overview of the temperature estimation errors associated with each model, allowing for a nuanced assessment of their accuracy

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and precision in capturing temperature variations. The LSTM model completed training in 5 min, and S CNN-LSTM and P CNN-LSTM required approximately 15.5 and 5 min for training, respectively; however, LSTM models begin with high error rates, which then sharply decrease, stabilizing around an error rate of approximately 1.2 in the process. This behavior indicates the adaptive nature of LSTM but also implies an initial instability that could be attributed to the model initialization parameters or learning curve of the model. In comparison to the LSTM model, the S CNN-LSTM model begins with a lower error rate and remains stable with minor fluctuations, which can be attributed to the convolutional layers' ability to capture spatial dependencies effectively. The P CNN-LSTM model exhibits similar behavior to the S CNN-LSTM model but with slightly higher stability; therefore, the P CNN-LSTM model may be more accurate in estimating battery STs.

With its memory retention capabilities, the LSTM network can effectively identify long-term patterns in sequential data. These long-term patterns are captured effectively by the LSTM network, as it gradually accumulates knowledge from previous cycles. In this way, the LSTM network refines its state estimation as it accumulates information from earlier cycles.

B. Validation Against Different Battery Chemistry Aged Under Driving Cycles: Dataset 2

This research established a broad prognostic framework aimed at predicting battery ST. The effectiveness of this method across various battery chemistries during dynamic operation is ensured by leveraging a thorough battery degradation dataset encompassing diverse degradation patterns. To further assess the applicability of the proposed method in EVs, experimentation was conducted using the Oxford battery degradation dataset, which incorporates different battery types and chemistries under dynamic discharge cycle conditions. In this section, we assessed the model's accuracy in predicting the behavior of batteries' ST as they aged under driving cycles at 40 °C. Analyzing the depicted ST of the batteries in Fig. 4 reveals noticeable fluctuations in cell temperatures over cycles as they age, even when subjected to consistent environmental conditions. Fig. 13 visually demonstrates all RMSEs and MAEs estimation of the three different models, LSTM, S CNN-LSTM, and P CNN-LSTM, across the battery's lifetime, which are also summarized in Table V. Fig. 14 demonstrates estimation results and their corresponding errors for some cycles, including the failure cycle (cycle#46) of test batteries, labeled as cell#5.

The LSTM model showcased moderate to commendable performance, as indicated by its RMSE values spanning from 0.72 to 0.95 and MAE values ranging between 0.67 and 0.90. These metrics suggest that the LSTM model yielded relatively accurate predictions. In contrast, the S CNN-LSTM model, which integrates convolutional layers in series with LSTM, exhibited less favorable outcomes. With RMSE values varying from 1.03 to 1.30 and MAE values ranging from 1.01 to 1.27, this model demonstrated higher error rates compared to the LSTM counterpart. The incorporation of additional convolutional layers appeared to introduce complexity without commensurate improvements in prediction accuracy. Conversely, the P CNN-LSTM model emerged as the most promising candidate among the three. With RMSE values spanning from 0.45 to 0.72 and MAE values ranging between 0.39 and 0.70, this architecture demonstrated superior predictive performance. The parallel arrangement of CNN and LSTM layers seemed to facilitate more effective feature extraction and learning, leading to enhanced prediction accuracy.

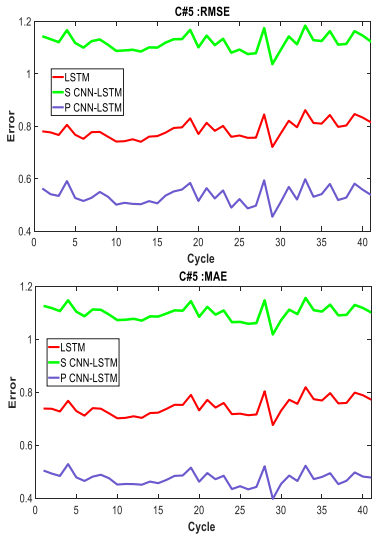


Fig. 13. RMSE and MAE of three models on Cell#5 subjected to 40 °C ambient temperatures.

TABLE V
RMSE AND MAE ST ESTIMATION ON CELL#5

Model	Cell#5	
	RMSE	MAE
LSTM	0.72 – 0.95	0.67 - 090
S CNN-LSTM	1.03 - 1.30	1.01 - 1.27
P CNN-LSTM	0.45 - 0.72	0.39- 0.70

The LSTM model completed training in 9.5 min, S CNN-LSTM and P CNN-LSTM required approximately 12 and 10.5 min for training, respectively. All these analysis underscores the critical role of model architecture in predictive modeling tasks. While the LSTM model exhibited commendable performance, the P CNN-LSTM architecture outshone both the LSTM and S CNN-LSTM models, emphasizing the importance of concurrent processing in capturing intricate data patterns. These findings offer valuable insights for researchers and practitioners seeking to optimize model selection for predictive tasks in various domains.

C. Validation Against Varying Ambient Temperature Under Driving Cycles: Dataset 3

Battery-powered EVs operate in real-world conditions with constantly fluctuating temperatures. In some instances, these EVs can encounter temperature changes of over 10 °C within a single day. It is, therefore, vital to assess models in scenarios that mimic these varying ambient temperatures, as represented by this dataset. During the training phase on the NCA cells

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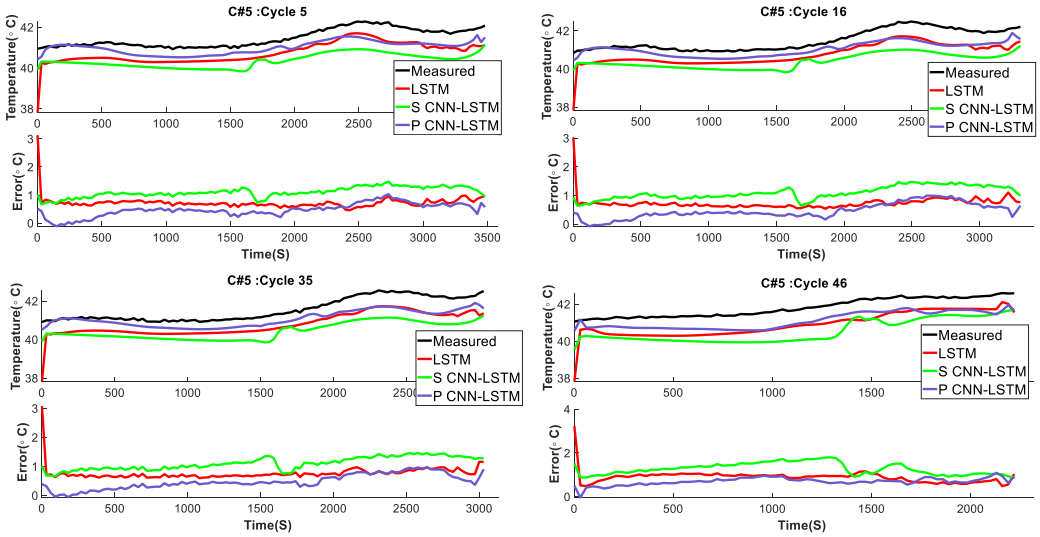


Fig. 14. ST estimation over cycles and its corresponding error for Cell#5 at a fixed temperature of 40 °C.

dataset obtained from the University of Wisconsin–Madison and McMaster University, our proposed models underwent a meticulous learning process. This involved training the models exclusively within a fixed range of ambient temperature data. The training set consisted of five fixed drive cycles, each meticulously conducted at a consistent ambient temperature. To validate the models, a separate dataset was constructed, comprising four standard drive cycles executed under stable ambient temperature conditions. To rigorously evaluate the models' adaptability to dynamic temperature conditions, we, however, conducted testing using mixed drive cycles that featured varying ambient temperatures, which are significantly different from the training data. These data are demonstrated in Table II as training, validation, and test data. This approach allowed us to assess the robustness and generalization capabilities of the proposed models in real-world scenarios where temperature fluctuations play a crucial role.

This validation experiment serves a dual purpose: it validates the effectiveness of the proposed model in adapting to diverse temperature environments and tests its compatibility with different battery materials. Fig. 15(a) and (b) illustrate the models' evaluation using dataset 3, and their corresponding error under varying ambient temperatures, starting at -20 °C and $+10$ °C, respectively. The comprehensive assessment of three distinct algorithms, LSTM, S CNN-LSTM, and P CNN-LSTM, in predicting battery ST across two rising ambient temperatures over four combinations of driving cycles are presented in Table VI. In addition to the tabular representation, the graphical depiction in Fig. 16 further enhances the understanding of the model performance, offering a quick comparison of the models' predictive capabilities. The results clearly show that the proposed methods outperform the LSTM model in all assessed criteria. The LSTM model completed training in 13.5 min, and S CNN-LSTM and P CNN-LSTM required approximately 15 and 14 min for training, respectively; however, it appears that the LSTM model may have difficulty handling the variability in the ambient temperature changes as a result of the fluctuating error rate. As the complexity of the data increases, the S CNN-LSTM model may become more stable but less accurate due to the general increase in error rates across cycles. The error rates, however, remain below the peaks of the LSTM model. A consistently lower error rate compared to the other two models indicates that PCNN-LSTM may be the most effective of the three models for predicting battery STs under varying ambient temperatures. Examining the RMSE and MAE ranges for each model provides valuable insights. LSTM exhibits stability, with RMSE ranging from 1.83 to 2.75 and MAE from 1.39 to 2.40, indicating consistent but slightly higher errors. In contrast, S CNN-LSTM stands out with its accuracy, showcasing a lower RMSE range of 0.88 to 1.85 and an MAE range of 0.72 to 1.64.

This suggests superior performance in capturing complex temperature dynamics. P CNN-LSTM proves to be a versatile contender, offering competitive results with RMSE 0.88 to 1.30 and MAE 0.68 to 1.12 ranges like S CNN-LSTM.

D. Discussion and Verification Against State-of-the-Art Studies

In summary, based on our comprehensive analysis of three datasets confirms the exceptional performance of hybrid models, S CNN-LSTM and P CNN-LSTM, surpassing the LSTM model in accurately estimating battery temperatures. While LSTM offers stable estimations, hybrid models exhibit superior accuracy and remarkable adaptability in dynamic conditions. Throughout the validation process across three distinct datasets, our proposed models consistently provide accurate estimations across various scenarios, including the battery's aging process, dynamic discharge cycles, and fluctuating ambient temperatures. Notably, P CNN-LSTM demonstrates the narrowest error ranges, enhancing accuracy by concurrently leveraging LSTM and CNN to extract spatial and temporal features independently of battery data. In selecting among these algorithms, it is essential to carefully consider the specific priorities of the application, striving for a balanced

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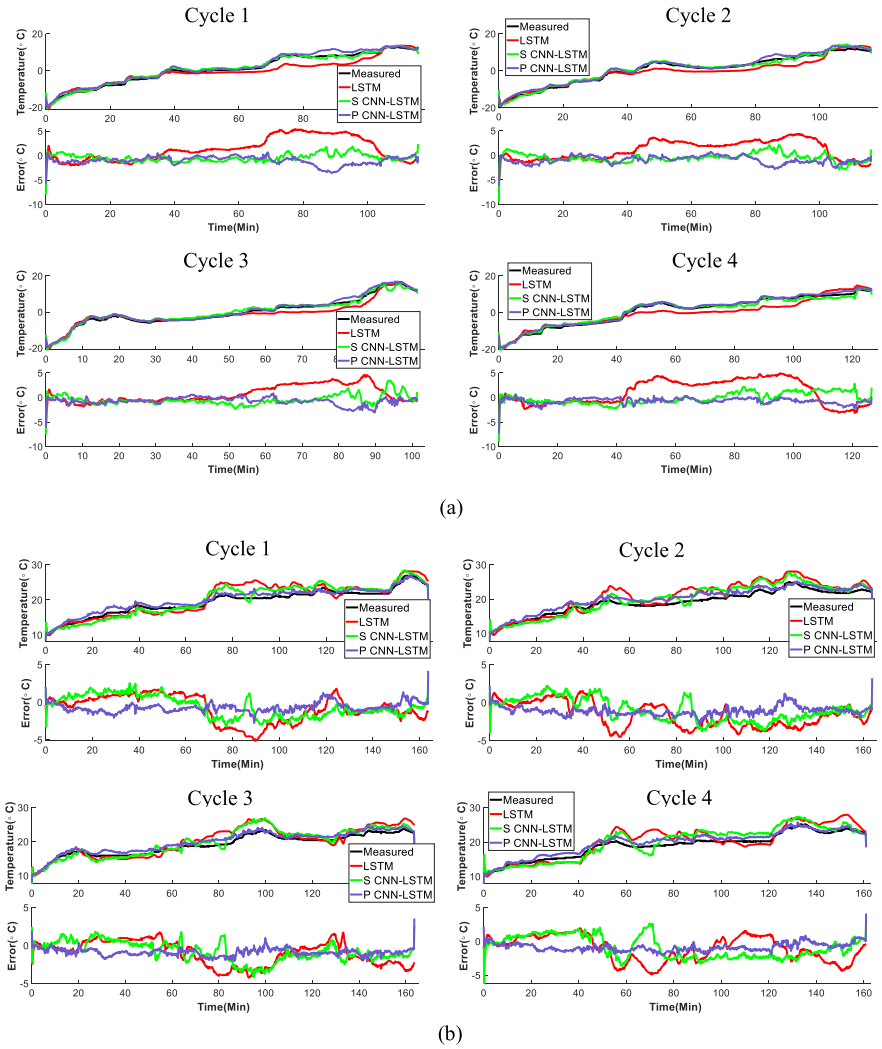


Fig. 15. (a) ST estimation and its correspondence error at varying ambient temperatures start at −20 °C and (b) +10 °C.

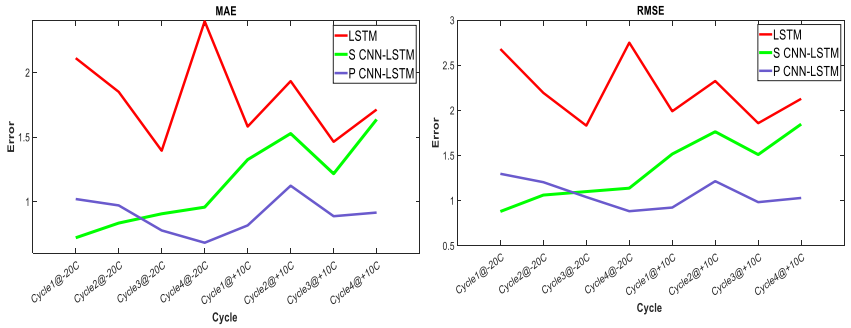


Fig. 16. RMSE and MAE of three models on batteries subjected to varying ambient temperatures.

approach that encompasses precision, adaptability, and stability in battery temperature estimation.

The proposed model validation against some state-of-the-art studies in battery surface estimation using different methods, [4], [7], [16], [32], are drawn in Table VII. A comprehensive comparison of different methods for battery temperature prediction reveals interesting insights. As part of the comparison, the type of algorithm, the ambient temperature at which the algorithms are deployed, and the associated errors are considered. While various models show varying levels of performance, the range of temperature and the associated prediction errors play a crucial role in determining their effectiveness. For instance, the ANN network demonstrates an impressively low error of less than 0.3 °C, which makes it suitable for precise temperature forecasts in a relatively stable ambient temperature condition of 24 °C. The experiment, however, only validated the different chemistries of batteries in new cells and aged cell health status. Limited adapt-

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TABLE VI
UNIVERSITY OF WISCONSIN–MADISON AND MCMASTER PREDICTION ERROR

Discharge Cycle	Varying Temperature at	Evaluation metrics					
		LSTM		S CNN-LSTM		P CNN-LSTM	
		RMSE	MAE	RMSE	MAE	RMSE	MAE
Cycle1	**-20°C**	2.68	2.11	0.88	0.72	1.30	1.02
Cycle2		2.19	1.85	1.06	0.83	1.20	0.97
Cycle3		1.83	1.39	1.10	0.91	1.04	0.78
Cycle4		2.75	2.40	1.14	0.96	0.88	0.68
Cycle1	**10°C**	1.99	1.58	1.52	1.33	0.92	0.82
Cycle2		2.32	1.93	1.76	1.53	1.21	1.12
Cycle3		1.86	1.46	1.51	1.22	0.98	0.89
Cycle4		2.13	1.71	1.85	1.64	1.03	0.92

TABLE VII
PROPOSED MODEL VERIFICATION AGAINST SOME STATE OF THE ART STUDIES

Model	Loading Profile	Fixed Ambient Temperatures	Varying Ambient Temperature	RMSE	MAE
ANN network [16]	Constant discharging	24°C	-	< 0.3 ° C	< 0.06 °C MAE
Impedance Measurement with Kalman Filter [4]	Artemis HEV drive cycle	8°C	-	< 1.24 °C	-
GRU-RNN network [7]	FUDS, US06 Combination of UDDS, LA92, US06, HWFET	-10°C, 0°C, 25°C, 50 °C	10 °C to 25°C	-	< 0. 20 °C (fixed T) < 0. 42 °C (varying T)
FNN LSTM [32]	UDDS, HWFET, US06, LA92, Mixed	-20 °C, -10 °C,0 °C,10 °C,25 °C	-20 °C to 40°C	< 2 °C < 1.8 °C	-
Proposed S CNN-LSTM	Constant discharging	43°C	-	< 0.97 °C	< 0.88 °C
	Artemis Urban drive cycle	40°C	-	< 1.30 °C	< 1.27 °C
	UDDS, HWFET, US06, LA92, Mixed	-	-20 °C to 15 °C 10 °C to 25 °C	< 1.85 °C	< 1.64 °C
Proposed P CNN-LSTM	Constant discharging	43°C	-	< 1.24 °C	< 1.08 °C
	Artemis Urban drive cycle	40°C	-	< 0.72 °C	< 0.70 °C
	UDDS, HWFET, US06, LA92, Mixed	-	-20 °C to 15 °C 10 °C to 25 °C	< 1.30 °C	< 1.02 °C

ability to varying ambient temperatures and current profiles, furthermore, makes it less suitable for real-world scenarios. It may not perform as well when dealing with broader temperature variations. On the other hand, the LSTM-RNN network demonstrated adaptability across a broader range, from −10 °C to 25 °C with an RMSE below 3 °C. This makes it highly adaptable to diverse ambient temperature conditions. The impedance measurement with Kalman Filter achieves remarkable accuracy with an RMSE of less than 1.24 °C, albeit within the limited temperature range of 8 °C. The GRU-RNN network offers a good performance with an MAE below 0.42 °C. The varying temperature condition, however, excludes the subzero temperature; it only experimented within 10 °C to 25 °C. The FNN and LSTM models exhibit solid performance within varied temperatures in the range of −20 °C to 40 °C, with an RMSE below 2 °C.

In contrast to the FNN and LSTM models, the proposed models operate effectively within the same range by exhibiting superior accuracy for identical dataset with RMSE values of less than 1.30 °C. This can prove the effectiveness of integrating CNN and LSTM components. Overall, the proposed S CNN-LSTM and P CNN-LSTM models provide robust performance across various operation conditions, emphasizing their suitability for accurate and adaptable battery temperature estimation. The proposed model presents a compelling option for battery temperature prediction, given its exceptional adaptability to a wide temperature range, encompassing dynamic and extreme conditions. This makes it particularly well suited for real-world applications, where temperature variations are significant and critical for battery performance and longevity.

VII. CONCLUSION

In summary, our research emphasizes the pivotal impact of temperature on LIB performance in EVs. We introduce an innovative deep learning network for precise battery ST estimation, leveraging BMS measurements through two hybrid models, S CNN-LSTM and P CNN-LSTM. These models exhibit exceptional performance and generalization

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across diverse scenarios, ensuring safe battery operation under varying conditions. The proposed method's effectiveness is thoroughly verified through comprehensive validation using three separate datasets characterized by different chemistries and operational conditions. This validation process includes testing the model under elevated ambient temperatures of 43 °C and 40 °C employing both CC and Artemis urban drive profiles. Additionally, the model is evaluated on batteries exposed to diverse, dynamic driving profiles across a spectrum of ambient temperatures ranging from 10 °C to 25 °C and from −20 °C to 10 °C. The accuracy of temperature estimation is assessed using RMSE, resulting in errors of less than 1.24 °C and 1.30 °C for fixed and varying ambient temperature conditions, respectively. These findings underscore the robustness and reliability of the proposed hybrid approaches in accurately estimating battery ST.

Comparative analysis with state-of-the-art studies, furthermore, highlights the performance of our proposed hybrid CNN-LSTM networks against alternative methods. It advances battery thermal management, emphasizing the efficiency and adaptability of ML in optimizing battery performance under dynamic operating conditions.

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